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Prof. Antonio Capone
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Subject: Letter of Support for the PRIN 2026 GridAI – GReen Itinerant Distributed AI Project

Dear Professor Capone,

the Institute of Power Transmission and High Voltage Technology at the University of Stuttgart, is pleased to express its support for the research project GridAI – GReen Itinerant Distributed AI, submitted by the consortium coordinated by Politecnico di Milano under the PRIN 2026 (Linea Principale) call of the Italian Ministry of University and Research.

The Institute of Power Transmission and High Voltage Technology at the University of Stuttgart is a leading German academic centre with expertise in power transmission and distribution system analysis, high-voltage technology, and electrical grid operation. The institute's research activities span grid planning and simulation, renewable energy integration, congestion management, residual-load pattern modelling, as well as flexibility aggregation and its grid-supporting application.

The subject of the project, namely the geographical and temporal mobility of artificial intelligence workloads in order to follow renewable energy availability and reduce the pressure on the electrical transmission infrastructure, lies in close continuity with our current research agenda on the integration of large flexible loads in renewable-dominated power systems.
A scientific collaboration with the GridAI consortium, in particular with the research group of Prof. Stefano Salsano at the University of Rome “Tor Vergata”, is currently active on the integration of computational load shifting with national-scale grid data and on the joint modelling of data-versus-energy trade-offs in the European energy transition. This collaboration is concretely anchored on the recent IFIP Networking 2026 paper by Baur, Paulus, Loh, Rudion and Hoßfeld on the integration of data centers and the German energy grid through geographic load shifting, in which our Institute contributed the energy-system modelling component.
We consider that the formulation proposed by the consortium, articulated around the concept of nomadic AI compute and the equivalence principle between electrical storage and acceleration of deferrable computation, represents a methodologically original direction with potentially high impact relative to the approaches currently prevailing at the European level, which are mainly focused on extending the HVDC backbone. The opportunity to extend this research line through a coordinated Italian-led consortium that adds an

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orchestration and trustworthy-certification angle is of significant value for the broader European conversation on renewable-native digital infrastructure.

We therefore confirm our interest in following the developments of the proposed project. In line with our existing collaboration and internal policies, we are open to exploring additional forms of technical and research cooperation, as well as the exchange of knowledge and data on topics of mutual interest during subsequent development phases.

Best regards,

K. Rudion

Prof. Dr.-Ing. habil. Krzysztof Rudion

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Stuttgart, 29.05.2026